

# # Heat and Thermodynamics

## • Heat

↳ Measure of the sum of kinetic energy of a substance.

↳ Heat energy is transferred from one body to another body as a result of difference in temperature.

↳ Heat flows from hotter body to the colder body.

↳ It is not necessary to flow heat from a body having more heat energy to a body having less heat energy.

↳ Heat energy depends upon both mass (no. of molecules present in a body) and temperature.

↳ SI unit of heat energy is Joule (J)

↳ Practical unit of heat energy is Calorie.

$$1 \text{ cal} = 4.2 \text{ J}$$

↳ Heat is measured by using a device calorimeter.



## • Temperature

- ↳ The degree of hotness or coldness of the body.
- ↳ Temperature of the body is the measure of average kinetic energy of molecules of a substance or body.
- ↳ Temperature determines the direction of heat flow from the body at higher temperature to body at a lower temperature.
- ↳ Device used to measure temperature is thermometer.
- ↳ SI unit of temperature is Kelvin (K). Also can be measured in Fahrenheit (°F) & degree Celsius (°C)

## • Effects of Heat

- ↳ change in Temperature
- ↳ change in Volume
- ↳ change in State of matter
- ↳ change in Physical properties
- ↳ change in chemical properties
- ↳ change in electric properties like resistance.



## • Difference between Heat and Temperature

| Heat                                                        | Temperature                                       |
|-------------------------------------------------------------|---------------------------------------------------|
| ↳ Sensation of hotness or coldness.                         | ↳ Degree of Hotness & coldness                    |
| ↳ S.I unit- Joule                                           | ↳ S.I unit- Kelvin                                |
| ↳ It is the cause                                           | ↳ It is the effect                                |
| ↳ Measured by calorimeter.                                  | ↳ Measured by Thermometer.                        |
| ↳ Sum of Kinetic energies of all its constituent particles. | ↳ Average kinetic energy of constituents of body. |

## • Thermal Equilibrium

↳ Two bodies or systems are said to be in thermal equilibrium if they are at same temperature and the net flow of heat between them is zero.

↳ It represents a state of equality of temperature between the bodies or systems.

↳ In this case there is no exchange of heat between two bodies in thermal con-



CLASSEMATE  
Date \_\_\_\_\_  
Page \_\_\_\_\_

tact. This concept forms the basis for a statement of zeroth law of thermodynamics.

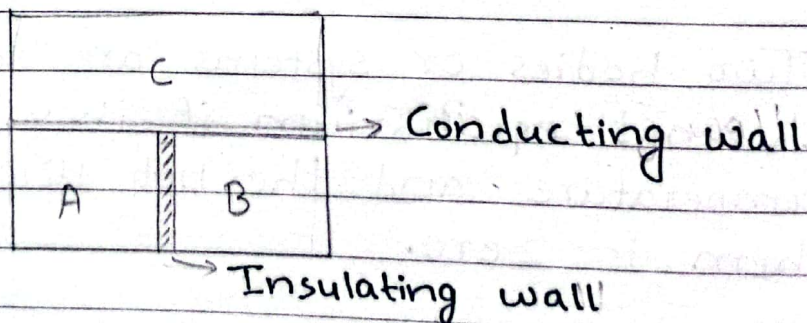
- For Thermal equilibrium to occur:

- ↳ The two bodies should be in thermal contact.

- ↳ Do not necessary to have same quantity of heat.

- Zeroth Law of Thermodynamics

- ↳ It states that "If two systems or bodies are separately in thermal equilibrium with the third system or body then they are also in thermal equilibrium with each other."



Consider, three thermodynamics systems A, B & C. Let, A & C are separated from each other by conducting wall, so they are in thermal equilibrium. Similarly, B & C are also separated from each other



by conducting wall, so they are also in thermal equilibrium. But, the systems A & B are separated from each other by insulated wall. Here, even though, wall between system A & B is insulated, they are also come in thermal equilibrium indirectly due to presence of third system.

If  $T_A$ ,  $T_B$  and  $T_C$  represents the temperature of systems A, B & C respectively.

When  $T_A = T_C$  &  $T_C = T_B$  Then,  $T_A = T_B$

So,

We have,  $T_A = T_B = T_C$

This clearly varifies the zeroth law of thermodynamics. Thus, it gives the basic concept of Temperature.

### • Thermometry

↳ The study of measurement of temperature of a body is called thermometry. The basic principle of thermometry is that the two bodies in thermal equilibrium have the same temperature.



## • Thermometer

↳ A device which is used to measure temperature of a body is called thermometer.

### ↳ Types of Thermometer

- Liquid Thermometers (Mercury thermometer and Alcohol thermometer)
- Gas thermometer (Hydrogen gas thermometers)
- Resistance thermometer (Platinum resistance thermometer)
- Thermoelectric thermometers
- Vapour pressure thermometers
- Radiation thermometer or Pyrometer

## • Different temperature scales

| S.N | Name       | L.B.P                | U.F.P                 | No. of division |
|-----|------------|----------------------|-----------------------|-----------------|
| 1.  | Centigrade | $0^{\circ}\text{C}$  | $100^{\circ}\text{C}$ | 100             |
| 2.  | Fahrenheit | $32^{\circ}\text{F}$ | $212^{\circ}\text{F}$ | 180             |
| 3.  | Kelvin     | $273\text{K}$        | $373\text{K}$         | 100             |
| 4.  | Reaumer    | $0^{\circ}\text{R}$  | $80^{\circ}\text{R}$  | 80              |



- Relation between different temperature scales.

$$\frac{C-0}{100-0} = \frac{F-32}{212-32} = \frac{K-273}{373-273} = \frac{R-0}{80-0}$$

$$\frac{C}{100} = \frac{F-32}{180} = \frac{K-273}{100} = \frac{R}{80}$$

$$\boxed{\frac{\Delta C}{100} = \frac{\Delta F}{180} = \frac{\Delta K}{100} = \frac{\Delta R}{80}}$$

- Faulty thermometer

$$\left( \frac{R-L}{U-L} \right)_{\text{correct}} = \left( \frac{R-L}{U-L} \right)_{\text{Faulty}}$$

- Numerical

1. A centigrade and a Fahrenheit thermometer are placed in hot water. The water is then cooled. What fall of temperature will the Fahrenheit thermometer register, when the centigrade thermometer records the fall in temperature as  $45^{\circ}\text{C}$ .

→ Soln,

$$\Delta C = 45^{\circ}$$

$$\Delta F = ?$$

We know that,



or,  $\frac{\Delta C}{5} = \frac{\Delta F}{9}$

or,  $\frac{45}{5} = \frac{\Delta F}{9}$

or,  $9 \times 9 = \Delta F$

$\therefore \Delta F = 81$

2. A faulty thermometer has its fixed points marked as  $5^\circ$  and  $95^\circ$ . Temp. of a body as measured the faulty thermometer is  $59^\circ$ . Find the correct temp. of the body on Celsius scale.

→ Soln,

faulty

L.P =  $5^\circ$

U.P =  $95^\circ$

$C_1 = 59^\circ$

Now,

$$\frac{C_1 - \text{L.P}}{\text{U.P} - \text{L.P}} = \frac{C_2 - 0}{100}$$

or,  $\frac{59 - 5}{95 - 5} = \frac{C}{100}$

or,  $C = \frac{54}{9} \times 100$

or,  $C = \frac{540}{9}$

$\therefore C = 60^\circ$



Date \_\_\_\_\_  
Page \_\_\_\_\_

## • Conceptual (Short) Questions.

1. What are the difference between heat and temperature.

→ The difference between heat and temperature is ÷

| Heat                                              | Temperature                                                             |
|---------------------------------------------------|-------------------------------------------------------------------------|
| → Sum of kinetic energy of the substance is heat. | → Measure of average of kinetic energy of the substance is Temperature. |
| → S.I unit of Heat is Joule                       | → S.I Unit of Temperature is Kelvin.                                    |
| → Heat is measured by calorimeter.                | → Temperature is measured by Thermometer.                               |

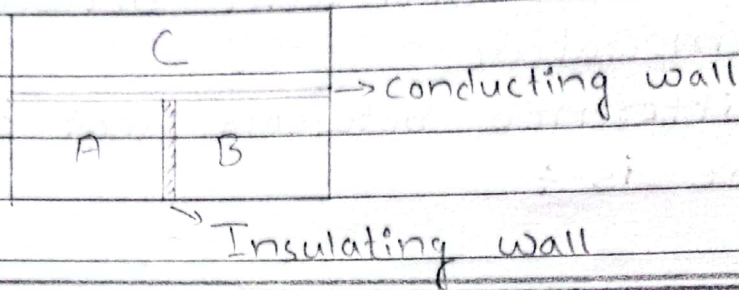
2. Define thermal equilibrium and explain Zeroth law of thermodynamics.

→ The condition in which temperature of two body is equal and net flow of heat between them is Zero, then it is called thermal equilibrium.

→ When two bodies or system A and B are in equilibrium with third body C then the two bodies A and B are



also in thermal equilibrium with each other.



3. A student claimed that thermometers are useless because a thermometer always registers its own temperature. How would you respond?

→ When a thermometer contact to our body physically the temperature ~~of~~ reading on the thermometer rises because it gains heat from our body until the temperature of both become equal. So, a thermometer always registers its own temperature.

4. Define absolute temperature.

→ The temperature measured in Kelvin scale (K) are called absolute temperature.

5. Express  $98^{\circ}\text{F}$  into Kelvin Scale

→ We know that,

$$\frac{F - 32}{180} = \frac{K - 273}{100}$$



$$\text{or, } \frac{98-32}{9} = \frac{K-273}{5}$$

$$\text{or, } \frac{66}{9} = \frac{K-273}{5}$$

$$\text{or, } \frac{22 \times 5}{3} = K-273$$

$$\text{or, } \frac{110}{3} + 273 = K$$

$$\text{or, } K = 309.66$$

$$\therefore 98^\circ \text{F} = 309.66 \text{ K}$$

6. At what temperature will the Celsius scale reading double the Fahrenheit reading?

→ Soln,

$$2F = C$$

We know that,

$$\frac{C}{5} = \frac{F-32}{9}$$

$$\text{or, } \frac{2F}{5} = \frac{F-32}{9}$$

$$\text{or, } 18F = 5F - 160$$

$$\text{or, } 18F - 5F = -160$$

$$\text{or, } F = \frac{-160}{13}$$

$$\text{or, } F = -12.30$$

$$\therefore F = -12.30$$

~~In Celsius,~~

$$\text{C} = 2F$$

$$\text{C} = 2 \times -12.30$$

$$\text{C} = -24.60$$



• Conceptual (Short) Questions.

1. At what temperature, do the Fahrenheit thermometer and Celsius thermometer show the same reading.

→ Soln,

$$\text{let, } C = F = x$$

$$\frac{C}{5} = \frac{F - 32}{9}$$

$$\text{or, } \frac{x}{5} = \frac{x - 32}{9}$$

$$\text{or, } 9x = 5x - 160$$

$$\text{or, } 9x - 5x = -160$$

$$\text{or, } 4x = -160$$

$$\text{or, } x = \frac{-160}{4}$$

$$\therefore x = -40$$

2. At what point of thermometric scale does Kelvin scale reading coincide with Fahrenheit scale reading?

→ Soln,

$$\text{let, } F = K = x$$

$$\frac{F - 32}{9} = \frac{K - 273}{5}$$

$$\text{or, } 5F - 160 = 9K - 2457$$

$$\text{or, } 5x - 160 = 9x - 2457$$

$$\text{or, } 9x - 5x = 2457 - 160$$

$$\text{or, } 4x = 2297$$



$$\text{or, } n = \frac{2297}{4}$$

$$\therefore n =$$

3. Why mercury is used in thermometer?  
→ Due to low specific heat capacity of mercury it can ~~easy~~ gain heat from other body fast and easily so, mercury is used in thermometer.

4. Calculate the change in temp in centigrade scale if the change in temp is Fahrenheit scale is  $45^\circ\text{F}$ .

→ Soln,

$$\Delta F = 45^\circ$$

$$\Delta C = ?$$

We know that,

$$\frac{\Delta C}{5} = \frac{\Delta F}{9}$$

$$\text{or, } \frac{\Delta C}{5} = \frac{45}{9}$$

$$\text{or, } \Delta C = 5 \times 5$$

$$\therefore \Delta C = 25^\circ$$

\* Which one of the following is the largest change in temperature?

a)  $1^\circ\text{C}$

c)  $1\text{K}$

b)  $1^\circ\text{F}$

~~d)  $1^\circ\text{R}$~~



Date \_\_\_\_\_  
Page \_\_\_\_\_

## Numerical & Figure Based Questions

1. The temperature of the melting point of ice and that of boiling point of water at normal atmospheric pressure are marked 40 and 200 respectively on a certain thermometer. What is

a) The temperature in  $^{\circ}\text{C}$  when the reading is  $80^{\circ}\text{C}$ .

→ Soln,

$$\frac{R-L}{U-L} = \frac{C-O}{100-O}$$

$$\text{or, } \frac{80-40}{200-40} = \frac{C}{100}$$

$$\text{or, } \frac{40}{160} = \frac{C}{100}$$

$$\text{or, } C = \frac{400}{160}$$

$$\therefore C = 25^{\circ}\text{C}.$$

b) The thermometer reading when the temperature is  $60^{\circ}\text{C}$ .

→ Soln,

$$\frac{R-L}{U-L} = \frac{C-O}{100}$$

$$\text{or, } \frac{R-40}{160} = \frac{60}{100}$$

$$\text{or, } R-40 = 96$$

$$\text{or, } R = 96 + 40$$

$$\therefore R = 136$$



2. A faulty Fahrenheit thermometer shows a temperature of  $206^{\circ}\text{F}$  when immersed in boiling water. When a correct Celsius thermometer is dipped in the same water it records  $96^{\circ}\text{C}$ . Find the correction to be applied to the faulty thermometer when it records  $206^{\circ}\text{F}$ .

→ Soln,

Correct reading =  $96^{\circ}\text{C}$

faulty reading =  $206^{\circ}\text{F}$

Now,

$$\frac{C - 0}{5} = \frac{F - 32}{9}$$

$$\text{or, } \frac{96}{5} \times 9 = F - 32$$

$$\text{or, } 172.8 + 32 = F$$

$$\text{or, } F = 204.8$$

$$\therefore F = 204.8^{\circ}$$

Again,

$$\begin{aligned}\text{fault} &= 204.8^{\circ}\text{F} - 206^{\circ}\text{F} \\ &= -1.2^{\circ}\text{F}\end{aligned}$$

$\therefore$  The correction is  $-1.2^{\circ}\text{F}$ . This means the faulty thermometer reads  $1.2^{\circ}\text{F}$  high, so we must subtract  $1.2^{\circ}\text{F}$  from its reading.

8122